BMW Price Prediction & Market Analysis

A data-driven study of resale pricing, depreciation, and predictive modeling insights

By: Naomi Onu
Tools used: Python and Power BI

Project overview

Objective:
Identify the key factors driving used BMW resale prices and evaluate if we can accurately predict a used BMW's price from its characteristics.

- Used BMW vehicle listings dataset sourced from Kaggle.
- 10,517 vehicle listings across 9 pricing and vehicle-related variables.

Methodology:

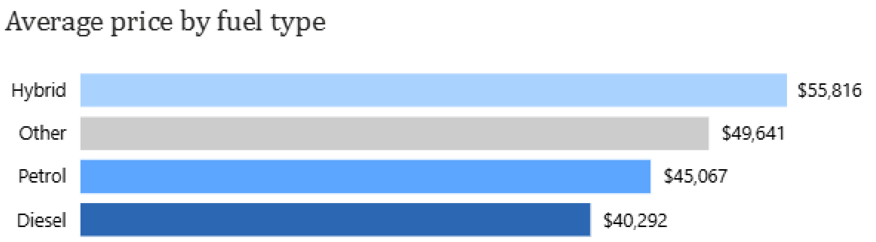
1. Data cleaning and preparation: removed outliers, engineered new variables (car age and price per km) and converted units to CAD and km.

2. Exploratory data analysis: built 8 visualizations in Python to identify patterns in pricing, depreciation, and market distribution.

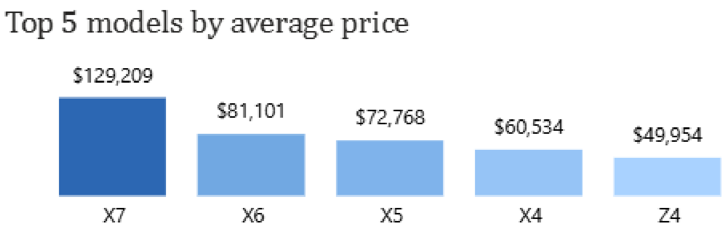
3. Statistical analysis: calculated correlation, analyzed price relationships across fuel type, transmission, and model.

4. Predictive modeling: trained and compared a linear regression and random forest model to predict resale price, achieving 90.6% accuracy.

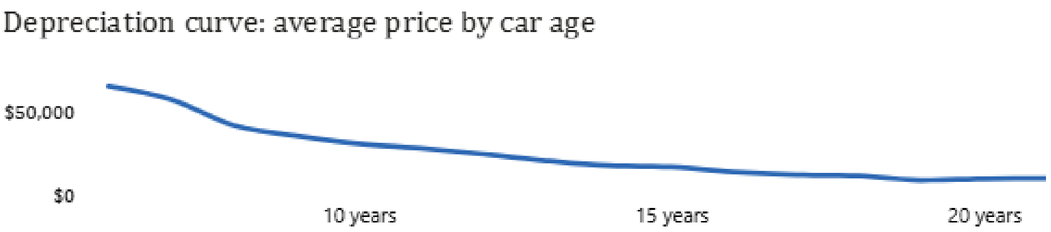
Key market findings



Hybrid cars command premium pricing



SUVs lead resale pricing

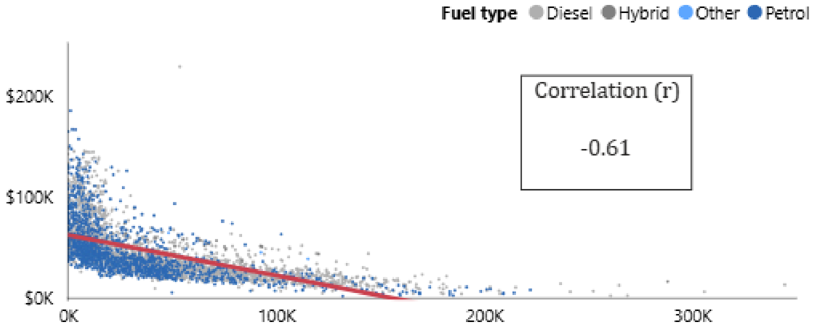


Depreciation is strongest during the first decade

Price drivers

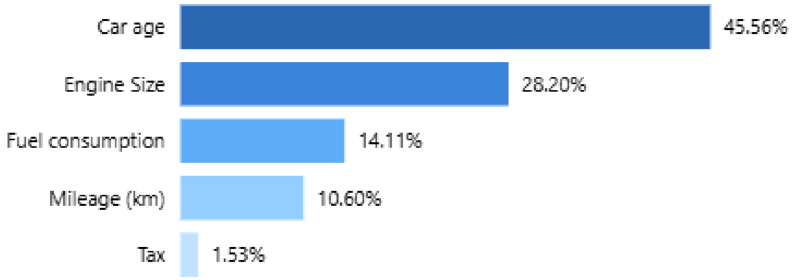
Exploratory analysis of the key factors influencing BMW resale pricing through correlation analysis and machine learning feature importance.

Price vs. mileage scatter by fuel type



Mileage has a moderately negative relationship with price

Key drivers of BMW resale value

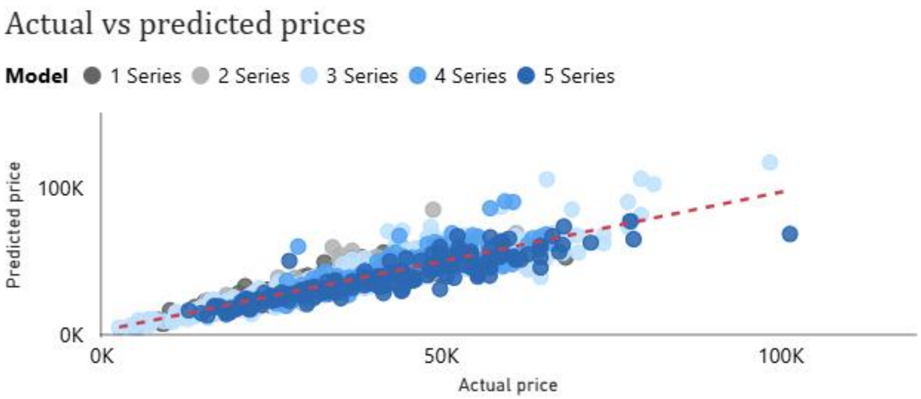


Car age accounts for 46% of price variance

Predictive model

The Random Forest model improved predictive performance by 14.6 percentage points in R^2 relative to linear regression.

Predictive performance (R^2 score)	Average prediction error (RMSE)	Linear regression R^2 (baseline model)	RF vs. linear improvement (R^2 gain)
90.60%	$6.49K	76.00%	14.60%
Strong fit	~15% of average price	Weaker fit	Significant



Executive takeaways

1. Vehicle age is the largest factor in resale price

   At 46% importance, the age of a BMW matters more than engine size, mileage, etc when determining its worth.

2. Used BMW prices sharply decline in the first 10 years.

   The depreciation curve shows the steepest drop before year 10, after which prices tend to stabilize.

3. SUVs have a strong value retention.

   X7, X6, X5, and X4 dominate the top 5 by average price, highlighting stronger long-term retention in the SUV segment.

4. Higher mileage has a moderately negative correlation with price.

   Age and model choice outweigh usage history in resale pricing.

5. Hybrid and electric BMWs demonstrate the highest average resale prices.

   Strong pricing potential as demand for alternative fuel vehicles continues to grow.

Thank you!

Questions and discussion

onun@mcmaster.ca
www.linkedin.com/in/naomi-onu
https://github.com/naomionu